

DD_CUST-INT-01 - Customer Interaction Timeline

Version: v1.0

Status: Draft for review

Bundle: 10_other

Depends on:

- DD_CROSS-00-System_Traceability_and_Implementation_Map-v1.0
- DD_API-00-API_Catalog_and_Gateway_Standards-v1.0
- FOUNDATION_AUTH
- FOUNDATION_KAFKA
- FOUNDATION_FILE_STORAGE
- DD_EM-CFG-03-RBAC_Catalog-v1.0
- DD_ILM-CFG-01-Customer_Service-v1.0
- DD_TCK-01-Ticketing_and_Case_Management-v1.0
- DD_NOT-01-Notification_Service-v1.3
- DD_SALES-01-Lead_Territory_and_Franchise_Sales_Management-v1.0

Future dependencies:

- DD_ASR-01-Technical_Trouble
- DD_ASR-02-Information_Request
- DD_ASR-03-Complaint
- DD_ASR-04-Service_Request
- DD_EM-03-CVM_and_Retention

1. Purpose

This DD defines the unified Customer Interaction Timeline for SOPHIX BSS.

The timeline gives customer care, supervisors, sales, service assurance, CVM, and reporting one chronological view of customer-facing activity:

- manual customer notes and structured interactions from ILM;
- ticket/case events from TCK;
- sales lead activity from SALES;
- customer notifications from NOT;
- work order/customer visit milestones from WO;
- billing/dunning customer-facing events from BIL;
- future ASR and CVM activities.

CUST-INT-01 is not a replacement for ILM, TCK, SALES, NOT, WO, or BIL. It is a normalized customer-history projection. Each owning module remains the source of truth for its own domain record.

2. What This Module Is For

Use CUST-INT-01 when the system needs to answer:

Question	CUST-INT answer
What happened recently with this customer across modules?	Query normalized timeline by customer/account/subscription.

Question	CUST-INT answer
Did customer care already contact this customer?	Show ILM interactions, ticket comments, notifications, sales activities.
What did the customer see or receive?	Show customer-visible notifications/comments/events.
What is internal-only context?	Show internal events only to authorized staff.
Can Customer 360 load one history panel?	Yes, from the timeline read API.
Can reporting count interactions by channel/type?	Yes, from normalized timeline event rows.

3. Scope

Included:

Capability	Purpose
Unified timeline projection	One read model across customer-facing modules.
Event ingestion	Consume Kafka events or internal API writes from source modules.
Normalization	Convert module-specific events into common timeline categories.
Visibility model	Separate internal-only, staff-visible, and customer-visible items.
Entity links	Link timeline event to ticket, lead, notification, invoice, WO, order, subscription, HomePass, or equipment.
Customer 360 read API	Provide filtered/paged history for Backoffice.
Self-care-safe read API	Provide only customer-visible timeline items if enabled.
Retention and masking	Support PII-safe summaries and retention policy.

Out of scope:

Out of scope	Owner
Customer/account master	ILM-CFG-01
Manual customer notes/interactions source of truth	ILM-CFG-01
Ticket/case lifecycle and ticket timeline	TCK-01
Sales lead/activity source of truth	SALES-01
Notification rendering/delivery and delivery log	NOT-01
Work order lifecycle and field notes	WO-01
Billing/payment/invoice facts	BIL modules
Formal audit log for all system mutations	Foundation/module audit logs
Data warehouse/reporting marts	REP-01 consumes this timeline if needed

4. Ownership Pattern

CUST-INT is a projection service.

Source event	Source of truth	CUST-INT behavior
Customer note appended	ILM	Store normalized timeline event referencing <code>note_id</code> .
Structured customer interaction	ILM	Store normalized interaction event referencing <code>interaction_id</code> .
Ticket created/resolved/commented	TCK	Store summarized ticket timeline event with <code>ticket_id</code> .
Notification dispatched/failed/suppressed	NOT	Store customer communication event with <code>notification_id</code> .
Sales lead activity	SALES	Store sales interaction summary with <code>lead_id</code> .
Work order appointment/finalized	WO	Store customer visit/service milestone with <code>work_order_id</code> .
Payment/dunning customer-facing action	BIL	Store billing communication or billing milestone summary.

Rule: if a user opens the item detail, Backoffice must call the owning module using the linked entity id. CUST-INT should not contain enough data to reconstruct or mutate the source entity.

5. Core Concepts

Concept	Meaning
Timeline event	A normalized customer-history item.
Source module	Module that owns the original record.
Source event id	Original event or record id used for idempotency.
Interaction category	Broad classification: note, call, ticket, notification, sales, visit, billing, system.
Visibility	Who may see the event: internal only, staff, customer.
Redacted summary	Safe short text displayed in Customer 360.
Entity link	Pointer to source detail such as ticket, invoice, WO, lead, notification.

6. Data Ownership

CUST-INT owns:

Entity	Purpose
customer_timeline_event	Normalized timeline row.
customer_timeline_link	Linked source/domain entities for the row.
customer_timeline_ingest_error	Failed ingestion records requiring review/retry.
customer_timeline_cursor	Consumer progress by Kafka topic/partition or source stream.

CUST-INT references:

Reference	Owner
customer_id, account_id	ILM-CFG-01
ticket_id	TCK-01
lead_id	SALES-01
notification_id	NOT-01
work_order_id	WO-01
invoice_id, payment_id, wallet_id	BIL modules
subscription_id	SUB-LM-01
homepass_id	RLM-CFG-01

7. Core Tables

7.1 customer_timeline_event

What it is for: stores one normalized customer-history item.

When to use it: Customer 360, customer-facing history where enabled, reporting counts, and cross-module support context.

Column	Type	Purpose	Sample
timeline_event_id	TEXT PK	Timeline row id.	cte_01J_CALL_001
operator_code	TEXT	Operator scope.	WIK
customer_id	TEXT	ILM customer id.	cust_01HX3K9M2P5
account_id	TEXT NULL	ILM account id when account-scoped.	acct_01HZ_KAREN
subscription_id	TEXT NULL	Subscription if event is service-scoped.	sub_01HZ_FIBER
event_time	TIMESTAMP	When the interaction happened in source module.	2026-05-29T13:15:00Z

Column	Type	Purpose	Sample
ingested_at	TIMESTAMP	When CUST-INT stored it.	2026-05-29T13:15:04Z
source_module	TEXT	Owning module.	TCK-01
source_event_id	TEXT	Source event/record id for idempotency.	evt_tck_01J_CREATED
source_record_type	TEXT	Source entity type.	TICKET
source_record_id	TEXT	Source entity id.	tck_01J2SUPPORT001
category	TEXT	NOTE, CALL, TICKET, NOTIFICATION, SALES, VISIT, BILLING, SYSTEM, CVM.	TICKET
event_type	TEXT	Specific event code.	TicketCreated
direction	TEXT	INBOUND, OUTBOUND, INTERNAL, SYSTEM.	INBOUND
channel	TEXT NULL	Contact/channel.	CALL_CENTER
title	TEXT	Short display title.	Technical support ticket opened
summary	TEXT	Redacted display summary.	Customer reported no internet.
visibility	TEXT	INTERNAL_ONLY, STAFF, CUSTOMER_VISIBLE.	STAFF
actor_user_id	TEXT NULL	Staff/system actor if known.	kc_user_csr_07
actor_display_name	TEXT NULL	Display name snapshot.	Jane CSR
customer_visible	BOOLEAN	Whether self-care can show it if enabled.	false
pii_level	TEXT	NONE, LOW, MEDIUM, HIGH.	LOW
metadata_json	JSONB	Safe structured metadata.	{"priority":"P2"}

Constraints:

- Unique (operator_code, source_module, source_event_id).
- summary must be redacted. Do not store raw ID numbers, passwords, full payment card data, or sensitive diagnostic blobs.
- customer_visible = true only if the source module marks the item customer-safe.

7.2 customer_timeline_link

What it is for: stores links from a timeline event to source/domain entities.

When to use it: Backoffice needs to open ticket detail, invoice detail, WO detail, or related customer data from a timeline row.

Column	Type	Purpose	Sample
timeline_link_id	TEXT PK	Link id.	ctl_01J_TCK_LINK
timeline_event_id	TEXT FK	Timeline row.	cte_01J_CALL_001
entity_type	TEXT	Linked entity type.	TICKET
entity_id	TEXT	Linked entity id.	tck_01J2SUPPORT001
link_role	TEXT	Why linked.	SOURCE_RECORD
display_ref	TEXT NULL	Human reference.	KE-TCK-2026-000001
created_at	TIMESTAMP	Link creation time.	2026-05-29T13:15:04Z

Allowed entity_type examples:

- CUSTOMER
- ACCOUNT
- SUBSCRIPTION
- TICKET
- WORK_ORDER

- INVOICE
- PAYMENT
- WALLET
- ORDER
- LEAD
- NOTIFICATION
- HOME_PASS
- EQUIPMENT_INSTANCE

7.3 customer_timeline_ingest_error

What it is for: stores failed ingestion attempts that need retry or repair.

When to use it: source event was malformed, customer id could not be resolved, visibility was invalid, or source event could not be normalized.

Column	Type	Purpose	Sample
ingest_error_id	TEXT PK	Error id.	ctie_01J_BAD_EVENT
operator_code	TEXT NULL	Operator if known.	WIK
source_module	TEXT	Source module.	NOT-01
source_event_id	TEXT	Source event id.	evt_not_01J_BAD
error_code	TEXT	Failure reason.	CUSTOMER_ID_MISSING
error_message	TEXT	Human repair detail.	Notification event did not include customerId.
payload_json	JSONB	Original payload or safe redacted payload.	{"eventType": "NotificationDispatched"}
status	TEXT	PENDING_RETRY, RESOLVED, IGNORED, FAILED_FINAL.	PENDING_RETRY
attempt_count	INT	Retry count.	1
created_at	TIMESTAMP	First failure time.	2026-05-29T13:20:00Z
updated_at	TIMESTAMP	Last update time.	2026-05-29T13:20:00Z

7.4 customer_timeline_cursor

What it is for: tracks ingestion progress for event consumers.

When to use it: Kafka/inbox consumers need replay protection and operational visibility.

Column	Type	Purpose	Sample
cursor_id	TEXT PK	Cursor id.	cur_tck_ticket_created_p0
source_name	TEXT	Topic/source stream.	sophix.ticket.events
partition_no	INT NULL	Kafka partition if applicable.	0
last_offset	BIGINT NULL	Last committed offset.	99102
last_event_time	TIMESTAMP NULL	Last event time processed.	2026-05-29T13:19:00Z
status	TEXT	ACTIVE, PAUSED, ERROR.	ACTIVE
updated_at	TIMESTAMP	Last cursor update.	2026-05-29T13:20:00Z

8. Event Normalization

Minimum mapping:

Source	Source event/record	Timeline category	Visibility default
ILM	CustomerNoteAppended	NOTE	STAFF
ILM	CustomerInteractionAppended	CALL or SYSTEM based on channel	STAFF

Source	Source event/record	Timeline category	Visibility default
TCK	TicketCreated	TICKET	STAFF or CUSTOMER_VISIBLE if source says customer-visible
TCK	TicketResolved	TICKET	STAFF or CUSTOMER_VISIBLE
TCK	ticket customer-visible comment	TICKET	CUSTOMER_VISIBLE
NOT	NotificationDispatched	NOTIFICATION	CUSTOMER_VISIBLE if message was customer-facing
NOT	NotificationFailed	NOTIFICATION	INTERNAL_ONLY
SALES	SalesLeadCreated	SALES	STAFF
SALES	SalesLeadConverted	SALES	STAFF
WO	WorkOrderAssigned	VISIT	STAFF
WO	WorkOrderFinalized	VISIT	STAFF or CUSTOMER_VISIBLE if source marks safe
BIL	dunning notice sent	BILLING	CUSTOMER_VISIBLE
BIL	manual write-off/adjustment	BILLING	INTERNAL_ONLY unless source marks customer-safe
CVM future	recovery/retention activity	CVM	STAFF

Rule: source modules should include safe display fields in events. CUST-INT should not call every source module synchronously during ingestion just to build a summary.

9. APIs

All APIs require Authorization: Bearer <jwt> or service token.

9.1 Read APIs

API	Purpose	Consumers
GET /api/customers/{customer_id}/timeline?limit=50	Customer timeline by customer.	Backoffice, customer service.
GET /api/accounts/{account_id}/timeline?limit=50	Account-scoped timeline.	Backoffice Customer 360.
GET /api/subscriptions/{subscription_id}/timeline?limit=50	Subscription/service-scoped timeline.	Backoffice subscription detail.
GET /api/customer-timeline/events/{timeline_event_id}	Timeline event detail.	Backoffice.
GET /api/customer-timeline/ingest-errors?status=PENDING_RETRY	Ingestion repair queue.	Platform ops.

Query parameters:

Parameter	Meaning	Example
from	Start timestamp.	2026-05-01T00:00:00Z
to	End timestamp.	2026-05-29T23:59:59Z
category	Filter category.	TICKET
visibility	Filter visibility.	CUSTOMER_VISIBLE
sourceModule	Filter source.	TCK-01
limit	Page size.	50
cursor	Pagination cursor.	eyJ0Ijo1...

9.2 Internal Ingestion API

Kafka is preferred. Internal API is allowed for modules that cannot publish events yet.

API	Purpose	Consumers
POST /internal/customer-timeline/events	Insert normalized or source event payload.	ILM/TCK/SALES/NOT/WO/BIL modules.

API	Purpose	Consumers
POST /internal/customer-timeline/events/bulk	Bulk backfill.	Migration/data repair.
POST /internal/customer-timeline/ingest-errors/{id}/retry	Retry failed ingestion.	Platform ops.
POST /internal/customer-timeline/ingest-errors/{id}/ignore	Mark failed event ignored with reason.	Platform ops.

Internal event sample:

```
{
  "operatorCode": "WIK",
  "customerId": "cust_01HX3K9M2P5",
  "accountId": "acct_01HZ_KAREN",
  "sourceModule": "TCK-01",
  "sourceEventId": "evt_tck_01J_CREATED",
  "sourceRecordType": "TICKET",
  "sourceRecordId": "tck_01J2SUPPORT001",
  "eventTime": "2026-05-29T13:15:00Z",
  "category": "TICKET",
  "eventType": "TicketCreated",
  "direction": "INBOUND",
  "channel": "CALL_CENTER",
  "title": "Technical support ticket opened",
  "summary": "Customer reported no internet.",
  "visibility": "STAFF",
  "customerVisible": false,
  "links": [
    {
      "entityType": "TICKET",
      "entityId": "tck_01J2SUPPORT001",
      "linkRole": "SOURCE_RECORD",
      "displayRef": "KE-TCK-2026-000001"
    }
  ]
}
```

Read response sample:

```
{
  "customerId": "cust_01HX3K9M2P5",
  "items": [
    {
      "timelineEventId": "cte_01J_CALL_001",
      "eventTime": "2026-05-29T13:15:00Z",
      "category": "TICKET",
      "eventType": "TicketCreated",
      "title": "Technical support ticket opened",
      "summary": "Customer reported no internet.",
      "visibility": "STAFF",
      "sourceModule": "TCK-01",
      "links": [
        {
          "entityType": "TICKET",
          "entityId": "tck_01J2SUPPORT001",
          "displayRef": "KE-TCK-2026-000001"
        }
      ]
    }
  ],
  "nextCursor": null
}
```

10. Business Rules

Rule	Description
CINT-1	Source modules remain source of truth. Timeline rows are read-model summaries.
CINT-2	Timeline ingestion must be idempotent by (operator_code, source_module, source_event_id).
CINT-3	Customer-visible timeline events must be explicitly marked customer-safe by source module.

Rule	Description
CINT-4	Internal-only notes, diagnostics, and staff comments must never appear in customer self-care.
CINT-5	If customer/account cannot be resolved, store an ingest error instead of dropping silently.
CINT-6	Timeline summary must be redacted before storage.
CINT-7	Entity links are references only; no cross-module foreign keys.
CINT-8	Reads must enforce RBAC and customer/account scope.
CINT-9	Corrections are additive. Do not mutate historical timeline content except for legal redaction/anonymization.
CINT-10	Customer 360 may tolerate missing timeline data; source modules still hold authoritative detail.

11. Frontend Usage

11.1 FE-APP-01 Backoffice

Backoffice uses CUST-INT for the Customer 360 history panel.

Area	API
Customer timeline panel	GET /api/customers/{customer_id}/timeline?limit=50
Account timeline panel	GET /api/accounts/{account_id}/timeline?limit=50
Filter by type	query category=TICKET, category=BILLING, etc.
Open source detail	Use timeline links[] and call owning module detail API.
Ingest error repair	Platform ops screen over ingest-error APIs.

UI rule: timeline is for scanning history. Source module detail pages remain the place to act.

11.2 FE-APP-04 Customer Self-Care

Self-care can use CUST-INT only for customer-visible events:

```
GET /api/customers/me/timeline?visibility=CUSTOMER_VISIBLE&limit=20
```

If this endpoint is not enabled, self-care continues to show ticket history from TCK and notification history from NOT separately.

12. Integration With Modules

Module	Integration
ILM-CFG-01	Emits notes/interactions events or CUST-INT consumes ILM events.
TCK-01	Emits ticket lifecycle/comment events for timeline.
SALES-01	Emits lead activity/conversion events for customer history after lead has customer/prospect link.
NOT-01	Emits notification dispatched/failed/suppressed events.
WO-01	Emits customer visit/work milestone events where customer-linked.
BIL	Emits customer-facing billing/dunning/payment events where needed.
ASR future	Emits service assurance flow events through TCK or directly if needed.
CVM future	Emits retention/recovery/upsell activities.
REP-01	Consumes timeline data for interaction reporting.

13. Infrastructure And Deployment

For a budget-conscious medium deployment:

Component	Recommendation
Runtime	Run inside ILM/customer-service pod or shared business-services pod. It can become separate later if event volume grows.
Database	PostgreSQL customer_timeline schema in shared business DB. Partition customer_timeline_event by month if volume grows.
Kafka	Consume module events through inbox/outbox pattern.
Cache	Optional Redis cache for first page of customer timeline, TTL 1-5 minutes.
Search	Start with PostgreSQL indexes. Add OpenSearch only if full-text timeline search becomes necessary.
Observability	Alert on ingest error backlog, consumer lag, and projection delay.

AWS recommendation:

Concern	AWS guidance
Compute	Shared ECS/EKS service initially.
DB	Existing RDS PostgreSQL with monthly partitions if needed.
Events	Existing MSK/Kafka.
Search	Avoid OpenSearch in first budget-conscious deployment unless operational search requires it.
Storage	Attachments remain in file storage; timeline stores only links.
Monitoring	CloudWatch/Prometheus alerts for lag and ingest errors.

14. Tests

Minimum tests:

Test	Expected result
Ingest valid TCK event	Timeline row and link created.
Ingest same source event twice	One row only, same response.
Ingest event missing customer id	Ingest error created.
Customer-visible false event	Not returned by self-care timeline.
Internal-only event	Returned to authorized staff only.
Customer timeline pagination	Stable ordering by event_time desc, timeline_event_id desc.
Link opens source module	Link contains enough info for frontend to call owning module.
RBAC missing customer scope	Timeline read returns 403.
Redaction test	Sensitive fields not stored in summary.
Consumer retry after transient failure	Event eventually ingested once.

15. Open Decisions For Review

Decision	Recommendation
Should CUST-INT own manual customer notes?	No. ILM already owns notes/interactions. CUST-INT projects them.
Should CUST-INT be a separate deployment?	Not initially. Start inside ILM/shared business services; split later if volume grows.
Should self-care show unified timeline?	Optional. Start with tickets/notifications separately unless product wants one customer-facing history.
Should timeline store full event payload?	No. Store safe summary and links; source module keeps full detail.
Should source modules publish events or call internal API?	Prefer Kafka events. Allow internal API for early modules until event coverage is complete.